

How to control Mayhem: The painted look of hair on TMNT

Brian Gossart

Mikros Animation, Canada
brian.gossart@mikrosanimation.com

Anne-Claire Leroux

Mikros Animation, Canada
anne-claire.leroux@mikrosanimation.com

Olivier Pierre

Mikros Animation, Canada
olivier.pierre@mikrosanimation.com

Gregory Coelho

Mikros Animation, Canada
gregory.coelho@mikrosanimation.com

ABSTRACT

TMNT Mutant Mayhem (2023) represented a great challenge artistically and technically. The director aimed for a look that would give the illusion that each frame would be hand-crafted 2D concept art. To break away from a digital or procedural feeling, we needed to approach the assets from modeling to surfacing in a unique way. This was especially challenging for the look and feel of the groom. To treat the groom in a way that it would feel cohesive with all the other objects and integrate seamlessly in the rendering scenes, we developed a layered system that combined modeled geometry, curves, a stylized surfacing, rendering, and lighting approach connecting the character work of modeling, grooming, surfacing, and lighting artists.

CCS CONCEPTS

• **Computing methodologies** → Animation.

KEYWORDS

Groom, hair, stylized look, rendering, surfacing, shading, character development

ACM Reference Format:

Brian Gossart, Olivier Pierre, Anne-Claire Leroux, and Gregory Coelho. 2024. How to control Mayhem: The painted look of hair on TMNT. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Talks (SIGGRAPH Talks '24)*, July 27–August 01, 2024, Denver, CO, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3641233.3664353>

1 INTRODUCTION

To achieve the desired hand-crafted look of the film, we established a set of standards for all assets. First, Modeling always needed to be imperfect, avoiding straight lines or clean curves. Second, pencil lines made of 3d curves were necessary to achieve the concept art appearance and were integrated in the building of all models. Third, the stylization of the render was achieved by the development of a custom shader that added a painterly look to each individual light in the scene.

Naturally, the approach for the character grooms needed to follow the same standards to integrate seamlessly. However, the

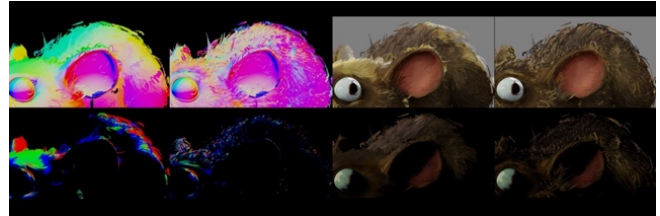


Figure 1: Normals override and its impact on thetoon shader.

painted look of hair and fur has been one of the most challenging parts of the characters, requiring us to mix different methods according to the nature of the different hair types as we outline in the following.

2 GROOMING: A LAYERED APPROACH

Before starting the grooming process, the character models were already extremely detailed, with faceting and imperfections carried by displacement maps, which already provided a lot of information to be considered in the grooming process, especially for characters with short hair such as Splinter. However, while modeling was a great first step to establish the look of graphic curves, it soon became apparent that exclusively using a geometry approach instead of hair for our hair look would not allow us to take advantage of the richness of the silhouette that a traditional grooming approach offered.

The look we pursued was a traditional 2D illustrator's approach who would not paint each single stroke of hair but would take shortcuts instead, by adding detail only in specific areas to suggest global shapes and indicate the way the light would catch strands of hair. Thus, we also needed to find a way to remove the micro details of each single hair on the surface. Additionally, the grooming department employed various strategies to accommodate different hair lengths and shapes, while respecting the ethnic quality using Peregrine Labs Yeti, as well as our in-house modeling base grooming system called Quiff.

We first started with a mix of classical procedural hair, propagated as a first layer. On top of that, we added an additional pass of graphic shapes. As a final layer we added more specific and designed shapes, enhancing the hand-crafted look of the groom. Throughout the entire pipeline, proxy geometries were used as hair deformer, but also to carry the hair graph and attribute information, compiling the whole result at render time with Quiff.

To achieve the desired look and keep the workflow optimized and manageable, both artistic and technical skills were required.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH Talks '24, July 27–August 01, 2024, Denver, CO, USA

© 2024 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0515-1/24/07

<https://doi.org/10.1145/3641233.3664353>



Figure 2: Human and Mutant example: Groom with regular surfacing (left), custom toon shader on groom (right).

3 FINAL LOOK: CUSTOM SHADER

For the final look, we first explored filters in post-production to simplify the details of the grooming, but we had concerns regarding the consistency and accuracy of the look together with the other assets. Because of that, we discarded this approach and looked for a surfacing solution that would give the desired look straight out of the rendering process, giving us more control to tweak the attributes.

One of the key elements was the customization of the normal vectors of the groom by overriding its native normal and detailed volume information with a simplified version. For short hair, this was inherited from the body mesh. For long hair, for example a ponytail, we had a specific way of overriding the groom normals by projecting those of the proxy geometry. Regarding the shapes we used as graphic brush strokes, we had to simplify the data along those curves, so that the intensity and color of the light would be the same on the whole shape. This strengthened the illusion of simple hand-drawn brush strokes. We combined this technique with the use of the `shadow_opacity` node of Arnold, both for the floating graphic curves and the rest of the groom. That way, we were able to avoid casting shadows for each individual hair.

The simplification of the volume and detail was necessary to make the tone mapping of our shader efficient from a graphic standpoint. The architecture of the hair toon shader required an adaptation from the shader used for geometries, since the render engine calculates the intensity of light on curves differently. We needed the hair and groom to react to the light the same way the geometry carrying them did, to get a believable result. We managed to adjust the hair shader so that the lighting department could light the hair the same way as other assets made of geometry.

With the given custom normal data, we were able to simplify the lighting information through the shader and apply the tonemap that created the stepping and color shifting in the falloff of the light. This was achieved through predefined setups from our lookdev department, but these setups were also tweakable according to the color keys by the lighting department in a shot context. Furthermore, compositing was able to have access to the same AOVs for both geometry and groom, to tweak the characters globally. Even though it has a very graphic look, the hair reacted accurately to the intensity and color of each light.

The breakup of the stepping of the tonemapping was enriched by the graphic quality of the shapes defined by grooming. It allowed for a very graphic and 2D like silhouette and rim light that would have been close to impossible to obtain with a PBR shader. Some assets were especially challenging, like young Splinter who had a long, twisted haircut, and required a lot of back and forth between the surfacing and grooming department. In some instances, we had to make significant modifications to achieve the 2D graphic look in lighting. For example, we needed to inflate the shape of the skull to visually reduce the length of the hair and make the geometry more adequate for the normal projection on the haircut.

4 CONCLUSION

To develop the layered concept efficiently, we had to work extremely collaboratively and form strong connections between the modeling, grooming, surfacing, and lighting department. It took several months until all teams were able to achieve the final look of Splinter, for example, get it approved, and find an effective workflow for all the grooms on this show. However, our final approach proved to be extremely beneficial as the resulting rendered images could almost be integrated as is and did not require compositing to make too many tweaks and cheats in post to achieve the intended graphic look. While our method still required certain workarounds in shading, our work on the movie gave us a useful guideline for future development that will ease the build of new 2D graphic looks for grooming at Mikros Animation moving forward.

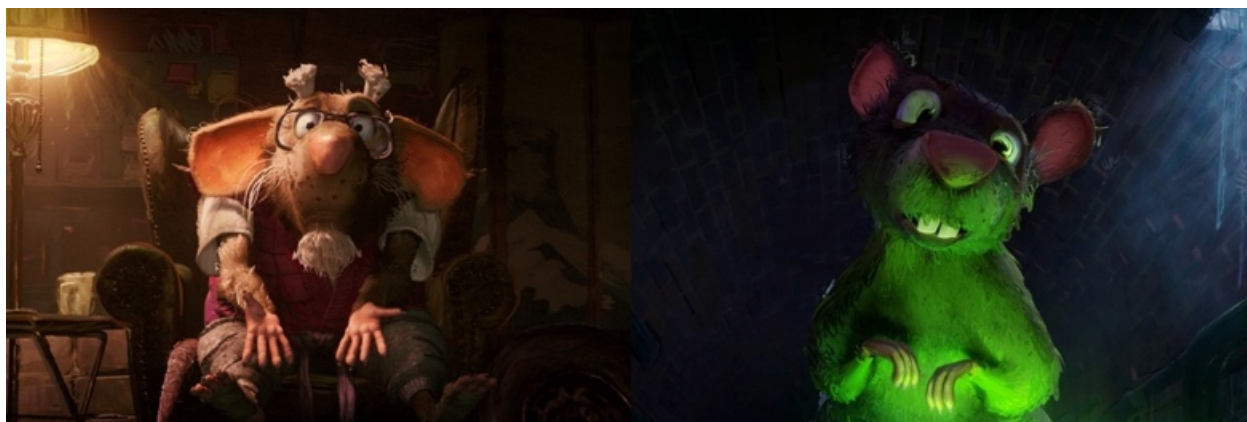


Figure 3: Final look of Splinter.